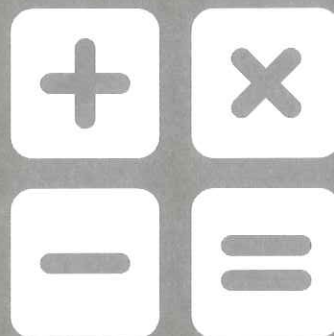




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER B



2010 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational
Assessment
Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

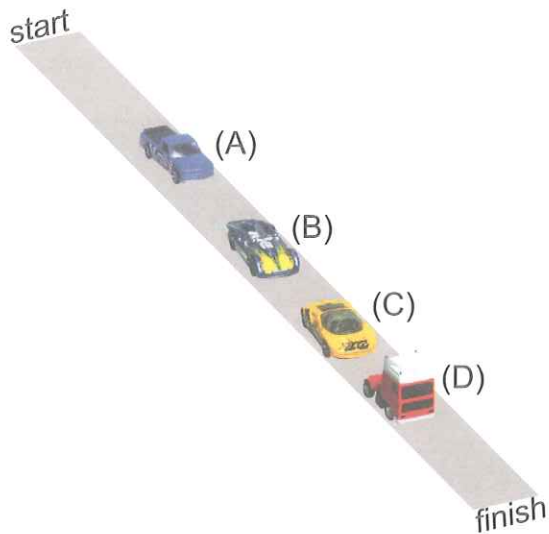
Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

1. Adam rolled four toy cars along a track.

Which car is about halfway along the track?



4. Adam's birthday is 8 July.

Sunil's birthday is exactly 3 weeks later.

JULY						
M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

What is the date of Sunil's birthday?

- (A) 20 July
(B) 22 July
(C) 27 July
(D) 29 July

2. This is a square centimetre.



How many square centimetres will cover this rectangle with no gaps or overlapping?



- (A) 6
(B) 8
(C) 10
(D) 12

3. $6 + 4 = \boxed{?} + 3$

- (A) 5
(B) 6
(C) 7
(D) 8

5. Manling has a bag of marbles. There are 10 green, 12 orange, 13 red and 15 yellow marbles.

Manling chooses one marble without looking.

Which colour is she most likely to choose?

- (A) red
(B) green
(C) yellow
(D) orange

HOW TO FILL OUT THIS SHEET:



- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

EXAMPLE 1: Debbie Bach

FIRST NAME

D	E	B	B	I	E
○	○	○	○	○	○
A	A	A	A	A	A
B	B	●	●	B	○
C	C	C	C	C	○
D	D	D	D	D	○

LAST NAME

B	A	C	H
○	○	○	○
A	●	A	A
●	B	B	○
C	C	C	○
D	D	D	○

EXAMPLE 2: Chan Ai Beng

FIRST NAME

C	H	A	N
O	O	O	O
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D

LAST NAME

A	I	B	E	N	G
O	O	O	O	O	O
A	A	A	A	A	A
B	B	B	B	B	B
C	C	C	C	C	C
D	D	D	D	D	D

EXAMPLE 3: Jamal bin Abbas

FIRST NAME	LAST NAME
J A M A L B I N	A B A S
☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐
☐ A ☐ A ☐ A ☐ A ☐ A ☐ A ☐ A	☐ ☐ A ☐ ☐ A
☐ B ☐ B ☐ B ☐ B ☐ B ☐ B ☐ B ☐ B	☐ B ☐ ☐ B ☐ ☐ B
☐ C ☐ C ☐ C ☐ C ☐ C ☐ C ☐ C ☐ C	☐ C ☐ C ☐ C
☐ D ☐ D ☐ D ☐ D ☐ D ☐ D ☐ D ☐ D	☐ D ☐ ☐ D

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / / Postcode:

DATE OF BIRTH

Day		Month		Year	
(0)	(0)	(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)	(1)	(1)
(2)	(2)		(2)	(2)	(2)
(3)	(3)		(3)	(3)	(3)
	(4)		(4)	(4)	(4)
	(5)		(5)	(5)	(5)
	(6)		(6)	(6)	(6)
	(7)		(7)	(7)	(7)
	(8)		(8)	(8)	(8)
	(9)		(9)	(9)	(9)

CLASS
(optional)

	(A)	(K)
	(B)	(L)
	(C)	(M)
	(D)	(N)
	(E)	(O)
	(F)	(P)
	(G)	(Q)
	(H)	(R)
	(I)	(S)
	(J)	(T)

TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: $4 + 6 =$

- (A) 2
- (B) 9
- (C) 10
- (D) 24

The answer is 10, so fill in the oval (C), as shown.

(A) (B) ☒ (C) (D)



START

1 (A) (B) (C) (D)

2 (A) (B) (C) (D)

3 (A) (B) (C) (D)

4 (A) (B) (C) (D)

5 (A) (B) (C) (D)

6 (A) (B) (C) (D)

7 (A) (B) (C) (D)

8 (A) (B) (C) (D)

9 (A) (B) (C) (D)

10 (A) (B) (C) (D)

11 (A) (B) (C) (D)

12 (A) (B) (C) (D)

13 (A) (B) (C) (D)

14 (A) (B) (C) (D)

15 (A) (B) (C) (D)

16 (A) (B) (C) (D)

17 (A) (B) (C) (D)

18 (A) (B) (C) (D)

19 (A) (B) (C) (D)

20 (A) (B) (C) (D)

21 (A) (B) (C) (D)

22 (A) (B) (C) (D)

23 (A) (B) (C) (D)

24 (A) (B) (C) (D)

25 (A) (B) (C) (D)

26 (A) (B) (C) (D)

27 (A) (B) (C) (D)

28 (A) (B) (C) (D)

29 (A) (B) (C) (D)

30 (A) (B) (C) (D)

31 (A) (B) (C) (D)

32 (A) (B) (C) (D)

33 (A) (B) (C) (D)

34 (A) (B) (C) (D)

35 (A) (B) (C) (D)

36 (A) (B) (C) (D)

37 (A) (B) (C) (D)

38 (A) (B) (C) (D)

39 (A) (B) (C) (D)

40 (A) (B) (C) (D)

6. Part of this picture is missing.



Which of these is the missing part?



(A)



(B)



(C)

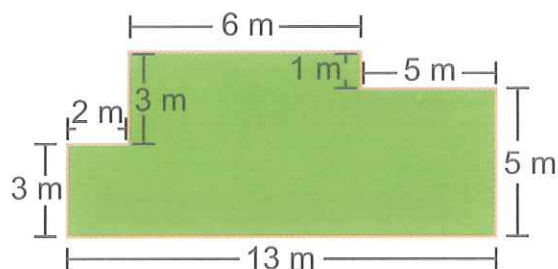


(D)

7. What is 3150 rounded to the nearest thousand?

- (A) 3000
- (B) 3100
- (C) 3200
- (D) 4000

8. Here is Noor's backyard.



Noor wants to fence her backyard.

How many metres of fence does she need?

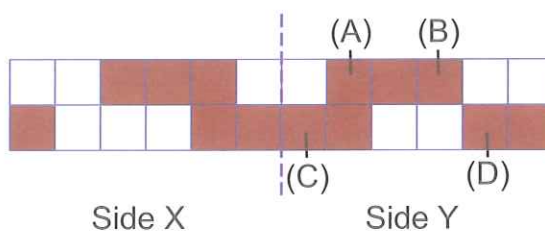
- (A) 27
- (B) 31
- (C) 35
- (D) 38

9. What is the missing number?

$$2 \times \boxed{?} + 4 = 10$$

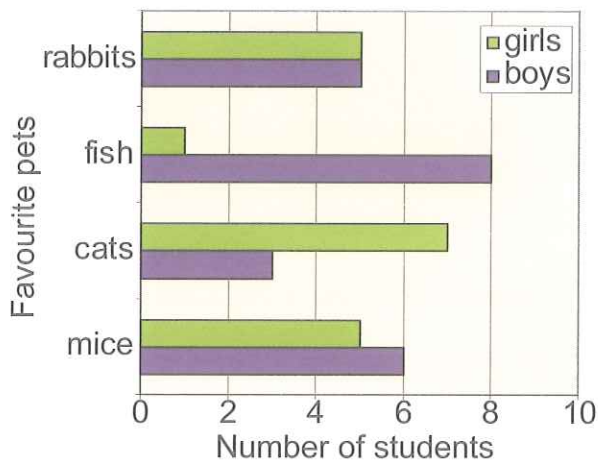
- (A) 3
- (B) 5
- (C) 6
- (D) 8

10. The bar should be shaded so that it is symmetrical.



Which square on Side Y should **NOT** be shaded?

11. Some students were asked to name their favourite type of pet.



Which type of pet was named by the greatest number of students?

- (A) rabbits
- (B) fish
- (C) cats
- (D) mice

14. This diagram shows how points are awarded for the first four places in each event at a school sports day.

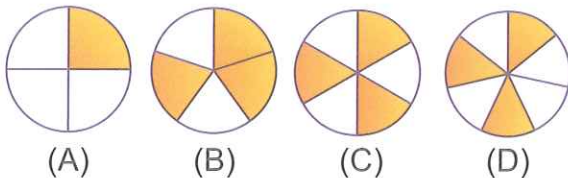


Tess went in 3 events and scored a total of 13 points.

What could her places have been?

- (A) 2nd, 3rd, 3rd
- (B) 1st, 3rd, 3rd
- (C) 2nd, 2nd, 4th
- (D) 1st, 3rd, 4th

12. Which circle has the largest area shaded?



15. Manling was facing south-west and then she turned 90 degrees clockwise.

Which direction was she then facing?

- (A) south-east
- (B) north-west
- (C) south
- (D) north

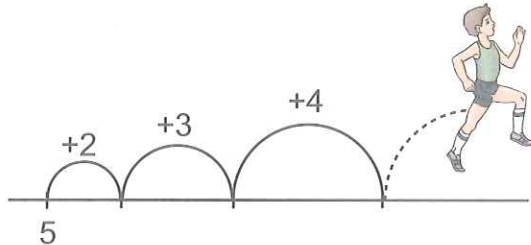
13. Luisa knows that $40 \times 14 = 560$. Using this information, she worked out 39×14 .

What calculation could Luisa have used to get the correct answer?

- (A) $560 - 40$
- (B) $560 - 39$
- (C) $560 - 14$
- (D) $560 - 1$

16. Adam is making a pattern by jumping on a number line and writing down the number he lands on.

He starts on the number 5 and jumps as shown in the diagram.



On what number will Adam land on his fourth jump?

- (A) 19
- (B) 18
- (C) 17
- (D) 14

17. A shop has 50 small plastic koalas like this one.

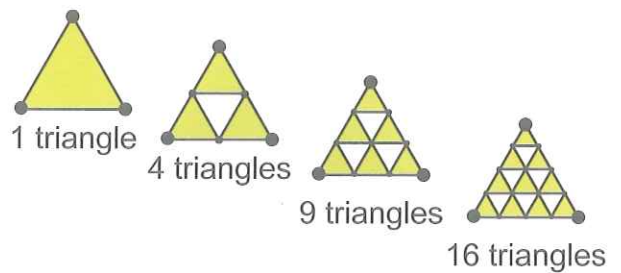


Each koala has a mass of 113 grams.

What is the total mass of all the koalas, in grams?

- (A) 5650
- (B) 5500
- (C) 565
- (D) 500

18.



How many triangles will be in the next shape of the pattern?

- (A) 25
- (B) 27
- (C) 30
- (D) 33

19. Noor is using this grid to represent the position of numbers.

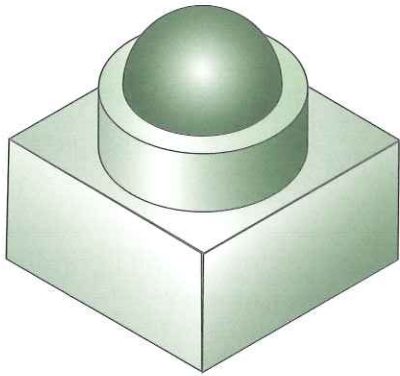
				
		3		
			5	
				
				

In this grid the position of the number 3 is given as (flower, tree).

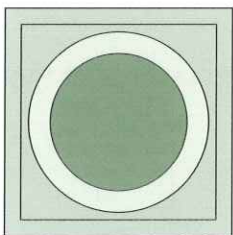
Using Noor's representation, what is the position of the number 5?

- (A) (sun, tree)
- (B) (tree, sun)
- (C) (boy, sun)
- (D) (sun, boy)

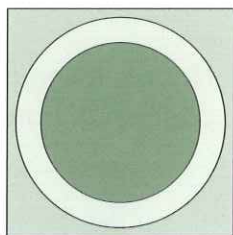
20. Jenny made this model of a building.



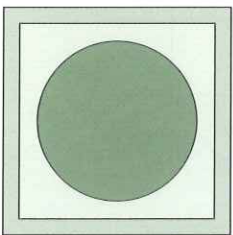
What does Jenny's building look like from above?



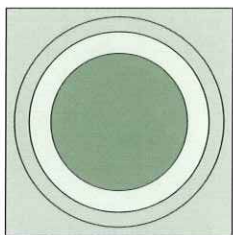
(A)



(B)



(C)



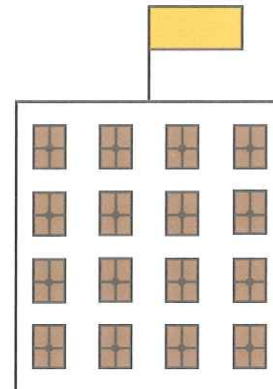
(D)

21. Sunil had a 20 m roll of masking tape. He cut off 15 cm.

How much tape is left on the roll, in cm?

- (A) 1985
(B) 500
(C) 185
(D) 5

22. This picture shows a building with 16 windows and a flag on top. The total area of the 16 windows is 32 square metres.



What is the area of the flag, in square metres?

- (A) 2
(B) 3
(C) 6
(D) 8

23. Adam has 24 chocolates and 72 sweets. He wants to use all of these to make identical birthday gift bags.

What is the largest number of gift bags he can make?

- (A) 72
(B) 24
(C) 12
(D) 8

24. This grid has 4 columns. The multiples of 5 are shaded and form a diagonal pattern on the grid.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	30	31	32
33	34	35	36
37	38	39	40

Luisa wants to make a grid where the multiples of 9 are shaded and form a diagonal pattern like this.

How many columns would Luisa need for her grid?

- (A) 5
(B) 6
(C) 7
(D) 8

25. Manling swaps ZING, ZANG and ZONG cards with a friend.

1 ZING card can be swapped for 3 ZANG cards.



1 ZANG card can be swapped for 4 ZONG cards.



Which group of cards has the highest value?



(A)



(B)



(C)



(D)

26. A class voted on whether they wanted a new school uniform.

This table shows the results.

	YES votes	NO votes
Boys	###	### ### I
Girls	### ###	###

What fraction of the girls did **not** want a new uniform?

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{5}{11}$ (D) $\frac{5}{31}$

28. Adam is hiding a 3-digit number. Here are a few hints to find out the number he is hiding:

- all the digits are odd numbers
- it can be divided by 5 with no remainder
- the difference between the hundreds digit and the tens digit is the same as the difference between the tens digit and the ones digit.

Which could be Adam's number?

- (A) 395
(B) 630
(C) 951
(D) 975

27. Luisa brought a basket of 30 small chocolate eggs to school.



She kept four eggs for herself.

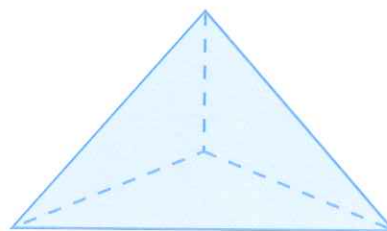
Then she gave two eggs to each of her friends.

She had six eggs left over.

To how many friends did Luisa give the eggs?

- (A) 20
(B) 16
(C) 10
(D) 7

29. The picture shows a triangular pyramid.



NOT TO SCALE

All the faces of this triangular pyramid are identical.

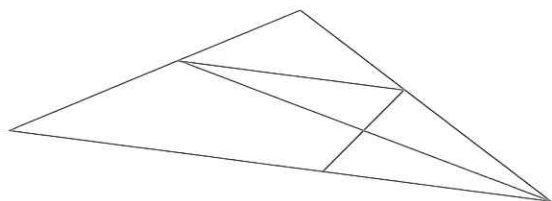
Noor has three of these triangular pyramids.

She glues the three pyramids together so that the joined faces overlap exactly.

How many faces does the new solid have?

- (A) 12
(B) 10
(C) 8
(D) 6

30. How many triangles are there in this shape?



- (A) 4
(B) 5
(C) 9
(D) 10

31. Sunil has these three number cards.

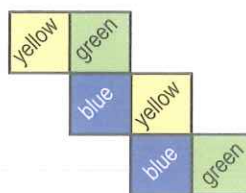


He picks each card in a random order and makes a 3-digit number. He uses each card only once.

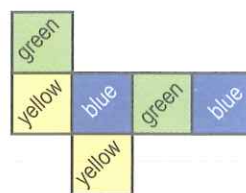
Which one of these does he have the greatest chance of making?

- (A) a number less than 400
(B) a number greater than 500
(C) a number between 150 and 450
(D) a number between 450 and 550

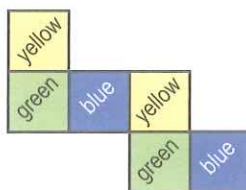
32. Which net would make a cube where no two faces of the same colour meet to form an edge?



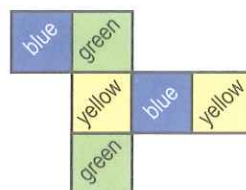
(A)



(B)



(C)



(D)

33. Adam trains by running up the stairs of a stadium.

With every stride he takes, Adam goes up three stairs. The stadium has 84 stairs.

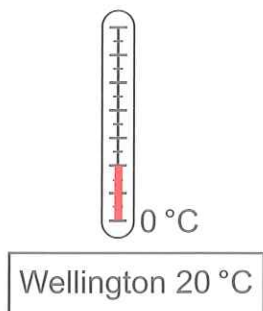
Adam is already halfway up.



How many more strides will he need to take to get to the top?

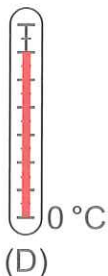
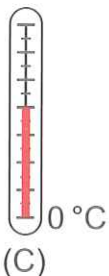
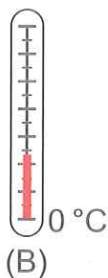
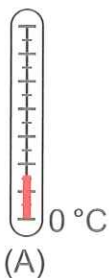
- (A) 8
(B) 14
(C) 28
(D) 126

34. This thermometer shows the temperature in Wellington. The numbers have been left off the scale.



The temperature in Brisbane is 4°C warmer than in Wellington.

Which thermometer shows the temperature in Brisbane?



36. Adam flew from Beijing to Melbourne. His flight left Beijing on Tuesday at 7:50 pm.

The total flight time was 13 hours and 20 minutes.

The time in Beijing is 2 hours behind Melbourne.

What was the local time in Melbourne when Adam arrived?

- (A) Wednesday 7:10 am
- (B) Wednesday 11:10 am
- (C) Wednesday 9:10 pm
- (D) Wednesday 11:10 pm

37. Manling needs 40 cm of ribbon to make one bow.



She has four rolls of ribbon, each one of a different colour. There are five metres of ribbon in each roll.

What is the largest number of single coloured bows Manling can make?

- (A) 52
- (B) 50
- (C) 48
- (D) 32

35. Half a litre of milk fills 2 cups.

How many cups can be filled with 3 litres of milk?

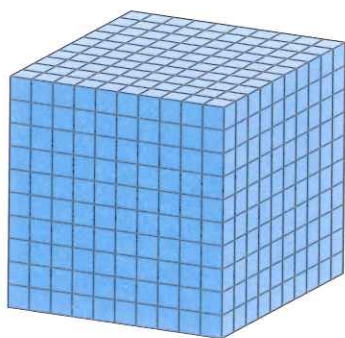
- (A) $1\frac{1}{2}$
- (B) 3
- (C) 6
- (D) 12

38. $3.12 + 14.98 =$?

- (A) 18.10
- (B) 18.00
- (C) 17.10
- (D) 17.00

39. Sunil had 1000 small white cubes.

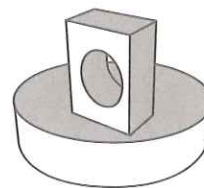
He glued them together to make this solid.
He then painted all faces of the solid blue.



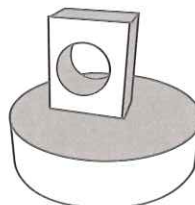
How many small cubes were **not** painted blue?

- (A) 640
- (B) 600
- (C) 512
- (D) 400

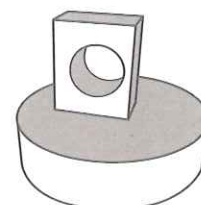
40. Luisa drew this object using her computer.



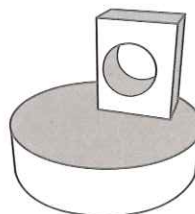
Which one of these is another drawing of the same object?



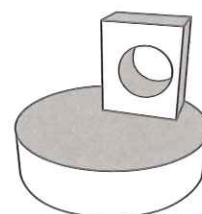
(A)



(B)



(C)



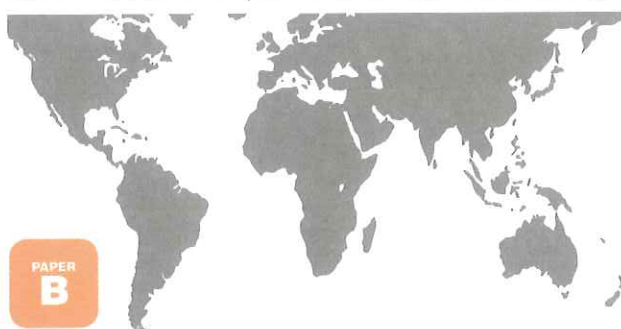
(D)

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Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific	Year 4
Singapore	Primary 3
South Africa	Grade 4



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Maths

	2010
1	B
2	C
3	C
4	D
5	C
6	B
7	A
8	D
9	A
10	D
11	D
12	B
13	C
14	B
15	B
16	A
17	A
18	A
19	D
20	B
21	A
22	C
23	B
24	D
25	D
26	D
27	C
28	D
29	C
30	D
31	A
32	D
33	B
34	B
35	D
36	B
37	C
38	A
39	C
40	A